Multi-Camera Person Re-Identification & Tracking

"One Identity.   Multiple Views.   Zero Confusion"

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1. Problem Statement

Develop a real-time system that can detect, track, and follow a person across multiple camera feeds. The system should allow an operator to select a person and maintain tracking continuity while enabling PTZ cameras to automatically follow the selected individual.

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2. Objective

The objective is to build a lightweight, real-time multi-camera tracking system that:

- Detects people in live RTSP streams
- Tracks multiple individuals per camera
- Maintains identity within each camera stream
- Allows manual selection of a target person
- Automatically controls PTZ cameras to follow the selected target

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3. System Overview

The system processes multiple camera feeds simultaneously and performs:

1. **Person Detection** using a deep learning model
2. **Multi-Object Tracking** within each camera
3. **User-Based Target Selection** via mouse input
4. **PTZ Camera Control** for automated tracking

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4. Key Features

4.1 Video Ingestion

- Supports multiple **RTSP camera streams**
- Each camera is processed in parallel using threaded frame readers
- Ensures low-latency frame access

4.2 Person Detection

- Uses **YOLOv8** model for detecting humans
- Filters detections based on confidence threshold
- Outputs bounding boxes for each detected person

4.3 Multi-Person Tracking

- Uses **DeepSORT** for tracking individuals
- Assigns unique track IDs per person
- Maintains identity across frames within a camera

4.4 Target Selection

- Operator can **click on a person** in the video window
- Selected person becomes the **active tracking target**
- System highlights the selected individual

4.5 PTZ Camera Control

- Supports ONVIF-compatible PTZ cameras
- Automatically adjusts camera pan/tilt to keep target centered
- Uses smoothing and dead-zone logic to:

    - Reduce jitter
    - Prevent unnecessary movement
    - Ensure stable tracking

4.6 Real-Time Visualization

- Displays camera feeds using OpenCV windows
- Shows bounding boxes and IDs
- Highlights selected target

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5. System Architecture (Simplified)

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RTSP Streams

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Frame Reader (Threaded)

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YOLO Detection

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DeepSORT Tracking

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Target Selection (User Click)

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PTZ Control Logic

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Display Output (OpenCV)
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6. Technology Stack

- **Python**
- **OpenCV** (video processing & display)
- **YOLOv8 (Ultralytics)** (person detection)
- **DeepSORT** (multi-object tracking)
- **ONVIF** (PTZ camera control)
- **NumPy** (numerical operations)

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7. Workflow

1. Capture frames from multiple RTSP cameras
2. Detect persons using YOLO
3. Track each person using DeepSORT
4. Display bounding boxes for all tracked individuals
5. User clicks on a person to select target
6. System continuously tracks selected person
7. PTZ camera adjusts position to follow the target

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8. Current Limitations

- No persistent identity across cameras (basic or limited ReID)
- No database or historical tracking
- No floor map visualization
- No alerting system
- PTZ follows only one person at a time

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9. Future Enhancements (Optional)

- Strong cross-camera ReID model (OSNet / FastReID)
- Multi-camera identity handoff
- Web-based dashboard
- Event logging and analytics
- Multi-target tracking with multiple PTZ cameras

What we achieved

This is NOT just a demo — it is:

- Real-time AI system
- Multi-camera capable
- PTZ-integrated (this is rare and impressive)
- Interactive (user-in-the-loop tracking)

References

- Demo Video
- Animated Presentation